

A rational contour proof of a Ramanujan-type $1/\pi^2$ identity

Hodge project: scalar Row12 manuscript

7 October 2026

Abstract

We evaluate a specific convergent weighted factorial series. The proof first converts the entire series into an angular and radial integral. An explicit rational coefficient one-form and a six-component rational correction give an antiderivative of the loaded radial integrand. Both endpoint limits and the removable interior singularity are treated explicitly. The nonzero endpoint is evaluated using an independently normalized Gauss Wronskian and a signed algebraic transformation. All finite rational inputs and an exact verification procedure are included in this single source file.

Version and verification scope. This manuscript concerns the literal scalar equality below. Its analytic derivation has evidence tags [ARG], and the retained exact rational checks have tag [ONE]. A fresh document-level review found no substantive defect in the scalar proof; its exact scope and source versions are recorded in `DOCUMENT_REVIEW.md`. Human mathematical acceptance is Pending. The literal scalar identity also has a completed Lean formalization as of 7 October 2026, including unconditional convergence and sum statements.

Contents

1	The statement and the proof structure	2
2	From the literal series to the loaded integral	2
3	The explicit rational coefficient form	3
4	The finite identities and the scalar primitive	4
5	The interior escape singularity is removable	6
6	The upper endpoint and the rational correction	6
6.1	The beta contour tends to zero at $U = 1$	6
6.2	All singular and constant terms of the outer correction cancel	6
7	The lower endpoint is a full positive residue	7
8	A native Gauss certificate for the $1/\pi$ value	8
8.1	The normalized symmetric square	8
8.2	A signed algebraic transformation	9
8.3	The Euler endpoint and the normalized Wronskian	11
8.4	The actual contact at $h = 64$	13
9	Completion of the series evaluation	13

A	A finite rational certificate for the source row	14
A.1	The complete source coefficient	14
A.2	Angular normal form and its six-state action	15
A.3	The outer primitive and all its coefficients	16
A.4	A complete exact verification recipe	17
B	Provenance and review boundary	21

1 The statement and the proof structure

Theorem 1.1. *The series on the left converges absolutely, and*

$$R := \sum_{n=0}^{\infty} \frac{(6n)!}{(n!)^6} \frac{532n^2 + 126n + 9}{10^{6n}} = \frac{375}{4\pi^2}. \quad (1)$$

Set

$$\rho = \frac{729}{15625} = \frac{6^6}{10^6}, \quad u = \frac{27}{125} = \sqrt{\rho}, \quad \tau = 2\pi i, \quad (2)$$

and introduce the two distinct weight polynomials

$$P(z) = 9 + 126z + 532z^2, \quad Q(z) = (1 + 6z)P(z) = 9 + 180z + 1288z^2 + 3192z^3. \quad (3)$$

The factor $1 + 6z$ in Q is essential: it comes from the radial moment. The argument constructs a function $\mathcal{E}$ satisfying

$$\mathcal{E}'(U) = -\frac{1}{3375} Q(\theta_\lambda) h(\rho(1 - U^2)^3), \quad \theta_\lambda = \lambda \frac{d}{d\lambda},$$

where $Q(\theta_\lambda)h$ is evaluated at the displayed argument. We prove, including every boundary contribution,

$$R = \int_0^1 Q(\theta_\lambda) h(\rho(1 - U^2)^3) dU, \quad \mathcal{E}(0+) = \frac{1}{36\pi^2}, \quad \mathcal{E}(1-) = 0.$$

The interior rational pole of the initially defined $\mathcal{E}$ is removable. These facts imply (1) by the fundamental theorem of calculus.

2 From the literal series to the loaded integral

For $n \geq 0$ define

$$a_n = \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} = \frac{(6n)!}{(3n)!(n!)^3 1728^n}, \quad (4)$$

where $(a)_0 = 1$ and $(a)_n = a(a+1) \cdots (a+n-1)$. The equality of the two formulas follows from their common value 1 at $n = 0$ and their common successive ratio

$$\frac{a_{n+1}}{a_n} = \frac{(n+1/6)(n+1/2)(n+5/6)}{(n+1)^3}. \quad (5)$$

In particular $0 < a_n \leq 1$. Define, on the complex unit disk,

$$F(q) = \sum_{n \geq 0} a_n q^n, \quad h(\lambda) = \sum_{n \geq 0} a_n^2 \lambda^n, \quad \theta = q \frac{d}{dq}. \quad (6)$$

The variable attached to each Euler operator will always be stated. All finite Euler derivatives converge normally on compact subsets of their unit disks, by the majorant $\sum n^j r^n$ for $r < 1$.

Lemma 2.1 (Radial moments). *For every integer $m \geq 0$,*

$$I_m := \int_0^1 (1 - U^2)^m dU = \frac{4^m (m!)^2}{(2m+1)!}.$$

Consequently

$$(6n+1)a_n^2 I_{3n} = \frac{(6n)!}{6^{6n}(n!)^6}. \quad (7)$$

Proof. The derivative of $U(1-U^2)^{m+1}$ has integral zero: the function vanishes at both endpoints. Its integral gives $(2m+3)I_{m+1} = (2m+2)I_m$. Starting at $I_0 = 1$ proves the stated formula. Substitution of (4) with $m = 3n$ proves (7); here $4^{3n}/1728^{2n} = 6^{-6n}$. $\square$

Absolute convergence of the original series follows independently from $(6n)!/(n!)^6 \leq 6^{6n}$, since this multinomial coefficient is one term in the expansion of $(1+\dots+1)^{6n}$ with six summands. Thus its terms are bounded by $P(n)\rho^n$, a summable sequence. Put $x = 1 - U^2$ and $\lambda = \rho x^3$. For $0 \leq U \leq 1$, the loaded summands are bounded by $Q(n)\rho^n$. This justifies termwise radial integration and yields the exact identity

$$R = \int_0^1 Q(\theta_\lambda) h(\rho(1-U^2)^3) dU. \quad (8)$$

Indeed the n th integrated term is $Q(n)a_n^2 \rho^n I_{3n}$, which equals the n th original term by (7). The term $n = 0$ has value 9 in both expressions.

Lemma 2.2 (Angular coefficient extraction). *For $0 < \lambda < 1$ and $\lambda < r < 1$, every positively oriented circle $|s| = r$ satisfies*

$$\theta_\lambda^j h(\lambda) = \frac{1}{\tau} \oint_{|s|=r} F(s) (\theta_b^j F)(\lambda/s) \frac{ds}{s}, \quad j \geq 0. \quad (9)$$

Here $\theta_b = b d/db$, and for $j = 0$ the factor is $F(b)$.

Proof. Both native power series and their required Euler derivatives converge absolutely and uniformly on the circle. Expanding the product, the normalized integral of $s^{m-n} ds/s$ equals 1 when $m = n$ and 0 otherwise. This leaves $\sum a_n^2 n^j \lambda^n$. For $j = 0$, the factor n^0 denotes 1, including at $n = 0$. The bounds already given also justify local parameter differentiation. $\square$

3 The explicit rational coefficient form

The coefficient recurrence gives the native differential equation

$$[\theta^3 - q(\theta + 1/6)(\theta + 1/2)(\theta + 5/6)]F = 0.$$

Writing $Y(q) = (F(q), \theta F(q), \theta^2 F(q))^T$, this becomes

$$\theta Y = N(q)Y, \quad N(q) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 5q & 23q & 3q \\ \hline 72(1-q) & 36(1-q) & 2(1-q) \end{pmatrix}. \quad (10)$$

Set

$$T(q) = \begin{pmatrix} -5q/72 & 0 & 2q^2(1-q) \\ -q/2 & q(q-1) & 0 \\ 1-q & 0 & 0 \end{pmatrix}, \quad W(q) = T(q)^T Y(q). \quad (11)$$

For use in the finite coefficient calculation, let

$$\mathbf{A}(q) = \begin{pmatrix} 0 & 1 & 0 \\ 2c(q) & -a(q) & 2 \\ 0 & c(q) & -2a(q) \end{pmatrix}, \quad a(q) = \frac{1-3q/2}{q(1-q)}, \quad c(q) = \frac{5}{144q(1-q)}. \quad (12)$$

Entrywise rational differentiation gives $T' + N^\top T/q + T\mathbf{A} = 0$. Hence

$$W'(q) = -\mathbf{A}(q)^\top W(q). \quad (13)$$

This identity fixes the dual-state convention used below.

For $0 < U < 1$ set

$$\begin{aligned} x &= 1 - U^2, & \lambda &= \rho x^3, & b &= \lambda/s, \\ m &= \frac{27x(80 - 99x + 27x^2)}{1000}, & \ell_0 &= \frac{27x(9x - 16)}{1000}, & \ell &= U\ell_0, \\ M &= 25 - 9x, & d &= x^4 M(99x - 50)^2, & f &= s^2 - 2ms + \lambda. \end{aligned} \quad (14)$$

The polynomial matrices $L_P(x), R_P(x)$ are explicitly specified in Appendix A. Define the rational matrix coefficients

$$B_s(U, s) = \frac{2UL_P(s - m) + 2\ell MR_P}{df}, \quad (15)$$

$$B_U(U, s) = \frac{UL_P \partial_U f - 2R_P\{\ell M \partial_U m + (s - m)(M \partial_U \ell + 9U\ell)\}}{df}. \quad (16)$$

The partial U derivatives hold s fixed. Both coefficients are used; the second summand in (15) is retained throughout. Introduce the actual normalized scalar contour function

$$\mathfrak{b}(U) = \frac{1}{\tau} \oint_{|s|=\sqrt{\lambda}} W(s)^\top B_s(U, s) W(\lambda/s) ds. \quad (17)$$

The polynomial identity $m^2 - \lambda = U^2 \ell_0^2 M$ gives the two roots

$$\sigma_\pm = m \pm U\ell_0 \sqrt{M}, \quad 0 < \sigma_+ < \sqrt{\lambda} < \sigma_-. \quad (18)$$

Indeed $m > 0$, $\ell_0 < 0$, $M > 0$ and $\sigma_+ \sigma_- = \lambda$ on $0 < x < 1$. The circle therefore contains exactly the inside root σ_+ and has no pole on it. The coefficients also have a rational pole at

$$99x - 50 = 0, \quad U_e = \frac{7}{3\sqrt{11}}. \quad (19)$$

For now (17) is used off this point.

4 The finite identities and the scalar primitive

This section makes explicit how finite rational identities imply an identity of actual analytic functions. Let $J = N(0)$, $\mathbf{w} = (5/72, 23/36, 3/2)$, and let E be the 3 by 3 matrix whose last row is $\mathbf{w}$ and whose other rows vanish. Write

$$\mathcal{M}_s = \frac{N(s) \otimes I - I \otimes N(\lambda/s)}{s}.$$

Flatten 3 by 3 matrices row by row; $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$ denote standard rows of length three. Define the six rows

$$\Psi_j = \frac{\mathbf{w} \otimes \mathbf{e}_j}{s - 1}, \quad \Psi_{j+3} = \frac{\mathbf{e}_j \otimes \mathbf{w}}{s - \lambda}, \quad 0 \leq j \leq 2. \quad (20)$$

The corresponding column of contour integrals is

$$v_j(\lambda) = \frac{1}{\tau} \oint_{|s|=r} \Psi_j(s, \lambda) [Y(s) \otimes Y(\lambda/s)] ds, \quad \lambda < r < 1. \quad (21)$$

These integrals are independent of the admissible radius, since their integrands are analytic in the intervening annulus. They are holomorphic in λ locally on the punctured unit disk by a fixed-circle argument: for complex λ choose $|\lambda| < r < 1$ in (21).

Lemma 4.1 (Finite rational certificate). *With the explicit constant matrix C , rational matrix $B_6(\lambda)$ and rational row $\xi(x)$ in Appendix A, one has*

$$C\mathbf{v}(\lambda) = (h, \theta_\lambda h, \dots, \theta_\lambda^5 h)^\top, \quad (22)$$

$$\theta_\lambda \mathbf{v} = B_6(\lambda) \mathbf{v}. \quad (23)$$

Define the two covariant coefficient derivatives

$$\begin{aligned} \mathcal{D}_U B_s &= \partial_U B_s + \frac{6U}{x} B_s b \mathbf{A}(b)^\top, \\ \mathcal{D}_s B_U &= \partial_s B_U - \mathbf{A}(s) B_U + \frac{b}{s} B_U \mathbf{A}(b)^\top, \end{aligned}$$

and $\mathbf{S} = T(s)(-\mathcal{D}_U B_s + \mathcal{D}_s B_U)T(b)^\top$. There is a rational row η with poles only at $0, 1, \lambda$ and infinity in s , such that

$$\text{flatten}(\mathbf{S}) = \sum_{j=0}^5 G_j(x) \Psi_j + \partial_s \eta + \eta \mathcal{M}_s. \quad (24)$$

Its six source coordinates satisfy the complete row identity

$$G - \frac{9c_0 + 180c_1 + 1288c_2 + 3192c_3}{3375} = \xi - 2(1-x)\xi' - \frac{6(1-x)}{x} \xi B_6(\rho x^3), \quad (25)$$

where c_j is row j of C (indices start at zero). All statements are rational identities on their regular charts.

Proof. Appendix A specifies every rational input and the finite elimination giving the primitive η . It verifies the anchor, the six parameter-action identities, and (25) using exact rational arithmetic. These identities, rather than a rank or numerical residual tolerance, are the finite certificate. For an exact row $\partial_s \zeta + \zeta \mathcal{M}_s$, its pairing with $Y(s) \otimes Y(\lambda/s)$ is the ordinary derivative of the single-valued function $\zeta[Y(s) \otimes Y(\lambda/s)]$. Its integral on the circle is zero. The anchor therefore gives $h = c_0 \mathbf{v}$, and the parameter-action identities give (23) and the subsequent five rows of (22). No regularity inside the circle is required for the integral of this total derivative to vanish. $\square$

For any $U_0 \in (0, 1) \setminus \{U_e\}$, root separation in (18) allows a fixed radius in a neighborhood of U_0 without crossing a pole of f . On this fixed compact circle all integrands and their derivatives are analytic in the parameter. Applying (13) and integrating the s derivative of the B_U pairing gives

$$\mathbf{b}'(U) = -G(x) \mathbf{v}(\lambda). \quad (26)$$

Here (24) supplies the source integral, with its displayed minus sign. The coefficient derivatives account for $\partial_U b = -6Ub/x$ and $\partial_s b = -b/s$.

By (23), ordinary differentiation gives

$$\frac{d}{dU} \{U \xi(x) \mathbf{v}(\lambda)\} = \left[\xi - 2(1-x)\xi' - \frac{6(1-x)}{x} \xi B_6 \right] \mathbf{v}.$$

Combining this with (25) and (26), define

$$\boxed{\mathcal{E}(U) = \mathfrak{b}(U) + U\xi(1 - U^2)\mathbf{v}(\rho(1 - U^2)^3)}. \quad (27)$$

It satisfies the actual function identity

$$\mathcal{E}'(U) = -\frac{1}{3375}Q(\theta_\lambda)h(\lambda), \quad \lambda = \rho(1 - U^2)^3. \quad (28)$$

5 The interior escape singularity is removable

At $x_e = 50/99$, direct substitution gives the two roots λ_e and 1 of f , with $0 < \lambda_e < 1$. Choose a fixed circle of radius $\lambda_e < r_e < 1$. For complex U sufficiently close to U_e , its integrands remain analytic on the circle. The continued integral and the rational correction in (27) are meromorphic in U ; their only possible local pole comes from powers of $99x - 50$. For real nearby U this circle is deformable to the balanced circle without crossing either root, so it represents the same function $\mathcal{E}$.

The right-hand side of (28) is holomorphic near U_e . Consequently $\mathcal{E}$ has no negative Laurent coefficient: the derivative of a nonzero term $a_{-k}(U - U_e)^{-k}$ would have a pole of order $k + 1$. Thus $\mathcal{E}$ extends holomorphically through U_e . Both real one-sided limits agree. This conclusion concerns the sum of the two terms in (27); every rational escape term in each coefficient remains included.

6 The upper endpoint and the rational correction

6.1 The beta contour tends to zero at $U = 1$

As $q \rightarrow 0$ in a disk, $F(q) = 1 + (5/72)q + O(q^2)$. Formula (11) shows that all three components of $W(q)$ are $O(q^2)$. In the first component the potentially linear terms are $-5q/72 + (5/72)q$ and cancel. The bounds are uniform on a sufficiently small disk.

As $U \rightarrow 1^-$, put $r = \sqrt{\lambda} = ux^{3/2}$. On $|s| = r$ both s and $b = \lambda/s$ have modulus r , so the product of the two dual-state bounds is $O(r^4) = O(x^6)$. The matrices L_P, R_P are bounded polynomials, $s - m = O(x)$ and $\ell_0 = O(x)$. Also

$$|f| \geq 2mr - 2\lambda = 2r(m - r) \geq c_1xr, \quad |d| \geq c_2x^4$$

for positive constants c_1, c_2 and sufficiently small $x > 0$. Indeed $m/x \rightarrow 54/25$ while $r/x \rightarrow 0$. It follows from (15) that $B_s = O(x^{-4}/r)$ uniformly on the circle. Including its length $2\pi r$ and the factor $1/\tau$,

$$\mathfrak{b}(U) = O(x^2) \rightarrow 0. \quad (29)$$

6.2 All singular and constant terms of the outer correction cancel

The literal matrix C gives

$$C^{-1}(1, 0, 0, 0, 0, 0)^T = v_0 = (0, 0, 0, 5/72, 0, 0)^T,$$

$$C^{-1}(1, 1, 1, 1, 1, 1)^T = v_1 = (-1, -1, -1, 77/24, 77/24, 77/24)^T.$$

Since $a_1^2\rho = 9/40000$, the normally convergent series give

$$\mathbf{v}(\rho x^3) = v_0 + \frac{9}{40000}x^3v_1 + O(x^6). \quad (30)$$

Expand the full rational row as $\xi(x) = \sum_{k=-4}^{\infty} \xi_k x^k$ near zero. Exact substitution of its printed coefficients yields

$$\xi_{-4}v_0 = \xi_{-3}v_0 = \xi_{-2}v_0 = 0, \quad (31)$$

$$\xi_{-1}v_0 = -\frac{105800}{459459}, \quad \frac{9}{40000}\xi_{-4}v_1 = \frac{105800}{459459}, \quad (32)$$

$$\xi_0v_0 = \frac{8464}{21879}, \quad \frac{9}{40000}\xi_{-3}v_1 = -\frac{8464}{21879}. \quad (33)$$

Both $(99x - 50)^{-1}$ and $(99x - 50)^{-2}$ contribute to ξ_0 . Equations (31)–(33) cancel all negative powers and the constant term of $\xi \mathbf{v}$. Multiplying the $O(x^6)$ remainder in (30) by $\xi = O(x^{-4})$ gives $O(x^2)$. Thus $\xi \mathbf{v} = O(x)$ and

$$\mathcal{E}(1-) = 0. \quad (34)$$

At the other endpoint $x \rightarrow 1$, the row ξ is finite and the state is analytic at $\lambda = \rho$. Consequently

$$U\xi(x)\mathbf{v}(\lambda) \rightarrow 0 \quad (U \rightarrow 0^+). \quad (35)$$

7 The lower endpoint is a full positive residue

Choose once and for all $\rho < r_0 < u$; for example $r_0 = 1/10$. For sufficiently small $U > 0$, both roots of f lie outside $|s| = r_0$, whereas the balanced circle encloses σ_+ alone. On the fixed inner circle $B_s \rightarrow 0$ uniformly: it has the explicit factor U , its remaining denominator is nonzero at $U = 0$, and both native arguments stay in compact subsets of the unit disk. Hence its normalized integral tends to zero.

The positively oriented annulus between the two circles contains exactly the pole σ_+ . The residue theorem gives their difference as the full residue of the scalar integrand in (17), because $2\pi i/\tau = 1$. Algebraically,

$$\text{Res}_{s=\sigma_+} B_s = \frac{UL_P + \sqrt{M}R_P}{d} \rightarrow S_* := \frac{4R_P(1)}{d(1)}. \quad (36)$$

No pole lies on either contour in this application. Taking the limit of the actual inside residue gives

$$\mathbf{b}(0+) = Y(u)^\top T(u) S_* T(u)^\top Y(u). \quad (37)$$

The polynomial matrices yield the small exact product

$$T(u) S_* T(u)^\top = \begin{pmatrix} -1/750 & 1/150 & 98/1125 \\ 1/150 & 98/1125 & 0 \\ 98/1125 & 0 & 0 \end{pmatrix}. \quad (38)$$

Every quantity in this computation is rational, except the subsequently evaluated native solution values. The sign is fixed by the annulus orientation and the selected inside root.

The native Gauss equation and symmetric-square identity proved in the next section imply, at $q = u$,

$$F = g^2, \quad \theta F = 2g\theta g, \quad \theta^2 F = 2(\theta g)^2 + 2g\theta^2 g,$$

with $\theta^2 g = (27/196)\theta g + (15/1568)g$ at u . Substitution in (38) gives

$$\mathbf{b}(0+) = \frac{\{3F(u) + 28\theta F(u)\}^2}{4500}. \quad (39)$$

For an explicit check, the coefficients of $g^4, g^3\theta g, g^2(\theta g)^2$ on both sides are respectively $9/4500, 336/4500, 3136/4500$.

8 A native Gauss certificate for the $1/\pi$ value

Put

$$\begin{aligned} H(p) &= \sum_{n \geq 0} \frac{(1/6)_n (5/6)_n}{(n!)^2} p^n, \\ F(q) &= \sum_{n \geq 0} \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q^n, \quad \theta = q \frac{d}{dq}, \quad \tau = 2\pi i. \end{aligned} \tag{40}$$

Here $(a)_0 = 1$ and $(a)_n = a(a+1) \cdots (a+n-1)$ for $n \geq 1$. Both series have radius of convergence one, by their consecutive coefficient ratios. Thus they and their termwise derivatives define analytic functions on the open unit disc. For real $0 < q < 1$, define

$$p(q) = \frac{1 - \sqrt{1-q}}{2}, \quad g(q) = H(p(q)), \quad Z(q) = H(1 - p(q)). \tag{41}$$

Every square root and fractional power of a positive real number below has its positive real value. A prime on H always means its ordinary derivative with respect to its argument.

8.1 The normalized symmetric square

The coefficient recurrence of H gives the actual Gauss equation

$$p(1-p)H''(p) + (1-2p)H'(p) - \frac{5}{36}H(p) = 0, \quad H(0) = 1. \tag{42}$$

Indeed its Euler form is $\{p\partial_p\}^2 H - p\{p\partial_p + 1/6\}\{p\partial_p + 5/6\}H = 0$, whose coefficient of p^n is $n^2 h_n - (n-5/6)(n-1/6)h_{n-1}$ for $n \geq 1$. The equation is invariant under $p \mapsto 1-p$, so $H(1-p)$ also satisfies it wherever $0 < p < 1$.

Write $s = \sqrt{1-q}$. The identities

$$p(1-p) = q/4, \quad 1-2p = s, \quad \frac{dp}{dq} = \frac{1}{4s}, \quad \frac{d^2 p}{dq^2} = \frac{1}{8s^3}$$

and the chain rule give, for either $y = g$ or $y = Z$,

$$\theta^2 y = \frac{q}{2(1-q)} \theta y + \frac{5q}{144(1-q)} y. \tag{43}$$

For clarity, the corresponding ordinary derivative equation is

$$y'' = \left(-\frac{1}{q} + \frac{1}{2(1-q)} \right) y' + \frac{5}{144q(1-q)} y,$$

where these primes mean q derivatives.

Define the two differential expressions

$$\begin{aligned} \mathcal{E}_2 y &= 144(1-q)\theta^2 y - 72q\theta y - 5qy, \\ \mathcal{E}_3 f &= 72(1-q)\theta^3 f - q(108\theta^2 f + 46\theta f + 5f). \end{aligned}$$

The product rule, without any transformation formula as an input, gives

$$\mathcal{E}_3(y^2) = y\theta(\mathcal{E}_2 y) + 3(\theta y)\mathcal{E}_2 y. \tag{44}$$

For example, the expansion of either side is

$$144(1-q)y\theta^3 y + 432(1-q)(\theta y)(\theta^2 y) - 216qy\theta^2 y - 216q(\theta y)^2 - 92qy\theta y - 5qy^2.$$

Consequently g^2 is annihilated by $\mathcal{E}_3$. The series F is annihilated by the same expression, since

$$n^3 f_n = (n - 5/6)(n - 1/2)(n - 1/6)f_{n-1} \quad (n \geq 1). \quad (45)$$

This recurrence uniquely determines every analytic solution from its constant coefficient: the multiplier n^3 is nonzero for $n \geq 1$. The function g is analytic near zero because $p(0) = 0$ and the square root in (41) is analytic there. Since $g(0)^2 = F(0) = 1$, the recurrence proves equality near zero. Both sides are real analytic on the connected interval $(0, 1)$, so

$$F(q) = g(q)^2 \quad (0 < q < 1). \quad (46)$$

This last continuation can equivalently be obtained by uniqueness for the nonsingular third-order ordinary equation on each compact subinterval of $(0, 1)$.

8.2 A signed algebraic transformation

For every real $h > 8$, set

$$\begin{aligned} v_1(h) &= \frac{(h-8)\sqrt{h+64}}{(h+16)^{3/2}}, & p_1(h) &= \frac{1-v_1(h)}{2}, & q_1(h) &= \frac{1728h}{(h+16)^3}, \\ v_2(h) &= \frac{(h-512)\sqrt{h+64}}{(h+256)^{3/2}}, & p_2(h) &= \frac{1-v_2(h)}{2}, & q_2(h) &= \frac{1728h^2}{(h+256)^3}, \\ L(h) &= \left(\frac{h+256}{h+16} \right)^{1/4}. \end{aligned} \quad (47)$$

We will prove the identity of actual functions

$$H(p_2(h)) = L(h)H(p_1(h)) \quad (h > 8). \quad (48)$$

The numerator $h - 512$ is signed throughout this assertion.

Here is a direct pullback and normalization proof, including the rational identities needed to check its gauge. Put $z = 1/h$ and

$$S = 1 + 64z, \quad d_1 = 1 + 16z, \quad d_2 = 1 + 256z, \quad \Delta = z \frac{d}{dz}.$$

The functions $\tilde{p}_i(z) = p_i(1/z)$ extend analytically to zero by the explicit formulas

$$\tilde{p}_1 = \frac{1 - (1-8z)\sqrt{S}/d_1^{3/2}}{2}, \quad \tilde{p}_2 = \frac{1 - (1-512z)\sqrt{S}/d_2^{3/2}}{2}. \quad (49)$$

The elementary polynomial identities

$$d_1^3 - (1-8z)^2 S = 1728z^2, \quad d_2^3 - (1-512z)^2 S = 1728z \quad (50)$$

imply

$$\tilde{p}_1(1 - \tilde{p}_1) = \frac{432z^2}{d_1^3}, \quad \tilde{p}_2(1 - \tilde{p}_2) = \frac{432z}{d_2^3}.$$

In particular $0 < \tilde{p}_i < 1$ when $0 < z < 1/8$; the first one is also less than $1/2$ because $1 - 8z > 0$.

Differentiating the literal signed expressions gives

$$\frac{d\tilde{p}_1}{dz} = \frac{864z}{\sqrt{S} d_1^{5/2}}, \quad \frac{d\tilde{p}_2}{dz} = \frac{432}{\sqrt{S} d_2^{5/2}}. \quad (51)$$

For an explicit check of these derivative numerators,

$$\begin{aligned} -8Sd_1 + (1-8z)(32d_1 - 24S) &= -1728z, \\ -512Sd_2 + (1-512z)(32d_2 - 384S) &= -864. \end{aligned}$$

Thus there is no extra factor $1 - 8z$ in either derivative. Their logarithmic derivatives on $0 < z < 1/8$ are

$$\frac{\tilde{p}_1''}{\tilde{p}_1'} = \frac{1}{z} - \frac{32}{S} - \frac{40}{d_1}, \quad \frac{\tilde{p}_2''}{\tilde{p}_2'} = -\frac{32}{S} - \frac{640}{d_2}, \quad (52)$$

where the derivatives in this display are with respect to z .

For any such pullback $Y(z) = H(\tilde{p}(z))$, the chain rule in (42) yields

$$\Delta^2 Y + B\Delta Y + CY = 0,$$

with the explicit coefficients

$$B = \frac{z\tilde{p}'(1-2\tilde{p})}{\tilde{p}(1-\tilde{p})} - 1 - \frac{z\tilde{p}''}{\tilde{p}'}, \quad C = -\frac{5}{36} \frac{z^2(\tilde{p}')^2}{\tilde{p}(1-\tilde{p})}. \quad (53)$$

All divisions here are legitimate by (50) and (51). Substitution gives

$$\begin{aligned} B_1 &= z \left(-\frac{8}{d_1} + \frac{32}{S} \right), & C_1 &= -\frac{240z^2}{Sd_1^2}, \\ B_2 &= z \left(-\frac{128}{d_2} + \frac{32}{S} \right), & C_2 &= -\frac{60z}{Sd_2^2}. \end{aligned} \quad (54)$$

For instance the first fractions in the expressions for B_1, B_2 in (53) are respectively $2(1-8z)/d_1$ and $(1-512z)/d_2$; together with (52) these produce exactly the displayed B_i .

Let

$$\mathcal{L}(z) = \left(\frac{d_2}{d_1} \right)^{1/4}, \quad \mu(z) = \frac{\mathcal{L}'(z)}{\mathcal{L}(z)} = \frac{60}{d_1 d_2}.$$

Its ordinary z derivative is

$$\mu' = -\frac{60(16d_2 + 256d_1)}{d_1^2 d_2^2} = -\frac{60(272 + 8192z)}{d_1^2 d_2^2}.$$

The gauge identities are

$$B_1 = B_2 + 2z\mu, \quad C_1 = C_2 + z^2(\mu' + \mu^2) + z(1 + B_2)\mu. \quad (55)$$

The first follows from $-8d_2 + 128d_1 = 120$. To verify the second without an implicit computational certificate, multiply it by $Sd_1^2 d_2^2$ and use this polynomial identity:

$$\begin{aligned} -240z^2 d_2^2 + 60z d_1^2 &= Sz^2(3600 - 960d_2 - 15360d_1) + 60z S d_1 d_2 \\ &\quad - 7680z^2 S d_1 + 1920z^2 d_1 d_2. \end{aligned} \quad (56)$$

It is an identity after $S = 1 + 64z$, $d_1 = 1 + 16z$ and $d_2 = 1 + 256z$ are inserted.

We also give the ordinary h equations explicitly. Since $\Delta = -h d/dh$ on a function pulled back by $z = 1/h$, the corrected coefficients (54) imply

$$Y_i'' + a_i Y_i' + b_i Y_i = 0, \quad Y_i(h) = H(p_i(h)), \quad (57)$$

where these primes are h derivatives and

$$\begin{aligned} a_1 &= \frac{1}{h} \left(1 + \frac{8}{h+16} - \frac{32}{h+64} \right), & b_1 &= -\frac{240}{h(h+64)(h+16)^2}, \\ a_2 &= \frac{1}{h} \left(1 + \frac{128}{h+256} - \frac{32}{h+64} \right), & b_2 &= -\frac{60}{(h+64)(h+256)^2}. \end{aligned} \quad (58)$$

Indeed $a_i = (1 - B_i(1/h))/h$ and $b_i = C_i(1/h)/h^2$. Writing

$$\mu_h(h) = \frac{L'(h)}{L(h)} = -\frac{60}{(h+16)(h+256)}, \quad \mu'_h(h) = \frac{60(2h+272)}{(h+16)^2(h+256)^2}, \quad (59)$$

the same identities become

$$a_1 = a_2 + 2\mu_h, \quad b_1 = b_2 + \mu'_h + \mu_h^2 + a_2\mu_h. \quad (60)$$

For the second conversion, explicitly $\mu_h = -\mu(1/h)/h^2$ and

$$\mu'_h + \mu_h^2 + a_2\mu_h = \frac{\mu'(1/h) + \mu(1/h)^2}{h^4} + \frac{(1 + B_2(1/h))\mu(1/h)}{h^3} = \frac{C_1(1/h) - C_2(1/h)}{h^2}.$$

Thus the product rule proves that LY_1 satisfies the second ordinary equation:

$$(LY_1)'' + a_2(LY_1)' + b_2LY_1 = L\{Y_1'' + (2\mu_h + a_2)Y_1' + (\mu'_h + \mu_h^2 + a_2\mu_h + b_2)Y_1\} = 0.$$

It remains to fix the actual solution, rather than only its equation. Both $H(\tilde{p}_2(z))$ and $\mathcal{L}(z)H(\tilde{p}_1(z))$ are analytic near zero, take the value one there, and satisfy $\Delta^2 Y + B_2 \Delta Y + C_2 Y = 0$; the displayed formulas give $\tilde{p}_1(0) = \tilde{p}_2(0) = 0$ and $\mathcal{L}(0) = 1$. The rational functions B_2, C_2 are analytic and vanish at zero. If their two solutions differed, the first nonzero term of their analytic difference would be cz^m with $m \geq 1$. The coefficient of z^m in its equation would then be $m^2 c$, because the other two terms have higher order. This contradicts $c \neq 0$. Hence the solutions agree near $z = 0$. On $h > 8$ the coefficients (58) are continuous and the equation is nonsingular. Ordinary initial-value uniqueness continues the equality from sufficiently large h to the entire connected interval $h > 8$, proving (48). This uniqueness step requires only the usual local uniqueness for a linear first-order system: on a compact interval its bounded coefficient matrix gives uniqueness by the integral equation and Gronwall's inequality.

All functions in (47) are smooth at $h = 512$. In particular $p_2(512) = 1/2$ and H is analytic there; the proof makes no change of branch at that point. Differentiating the actual identity now gives the fully explicit ordinary derivative identity

$$-\frac{432h H'(p_2(h))}{\sqrt{h+64}(h+256)^{5/2}} = L(h) \left\{ -\frac{60H(p_1(h))}{(h+16)(h+256)} - \frac{864H'(p_1(h))}{\sqrt{h+64}(h+16)^{5/2}} \right\}. \quad (61)$$

Here the two chain derivatives are exactly $p'_1(h) = -864/[\sqrt{h+64}(h+16)^{5/2}]$ and $p'_2(h) = -432h/[\sqrt{h+64}(h+256)^{5/2}]$.

8.3 The Euler endpoint and the normalized Wronskian

We derive the normalization from an integral, including its differentiated endpoint. For $x, y > 0$ write $B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$. The change of variable $t = r/(1+r)$ and then $r = w^6$ gives

$$B(5/6, 1/6) = \int_0^\infty \frac{r^{-1/6}}{1+r} dr = 6 \int_0^\infty \frac{w^4}{1+w^6} dw.$$

Here is an elementary residue evaluation of the last integral. The upper half-plane poles of $w^4/(1+w^6)$ are $\exp(i\pi/6), i, \exp(5i\pi/6)$, and the residue at a pole ζ is $1/(6\zeta)$. Their sum is

$$\frac{1}{6} \{e^{-i\pi/6} + e^{-i\pi/2} + e^{-5i\pi/6}\} = -\frac{i}{3}.$$

For a semicircle of radius $R \geq 2$ the absolute value of the integrand is at most $2/R^2$, so its arc integral tends to zero. The residue theorem therefore gives $\int_{-\infty}^\infty w^4/(1+w^6) dw = 2\pi/3$. The integrand is even, and consequently

$$B(5/6, 1/6) = 2\pi. \quad (62)$$

For $0 \leq x < 1$, binomial expansion in the positive integral yields

$$H(x) = \frac{1}{2\pi} \int_0^1 t^{-1/6} (1-t)^{-5/6} (1-xt)^{-1/6} dt. \quad (63)$$

To see the coefficients without assuming an Euler formula, integration by parts gives $B(a+1, b) = aB(a, b)/(a+b)$ for $a, b > 0$. Indeed the integral of $(t^a(1-t)^b)'$ is zero, and $B(a, b) = B(a+1, b) + B(a, b+1)$. It follows that $B(5/6+n, 1/6)/B(5/6, 1/6) = (5/6)_n/n!$. Inserting the binomial coefficients $(1/6)_n/n!$ in the integral therefore produces exactly (40); monotone convergence justifies the interchange for $0 \leq x < 1$.

On every compact interval $0 \leq x \leq r < 1$, differentiation under this integral is justified by the integrable majorant

$$\frac{(1-r)^{-7/6}}{12\pi} t^{5/6} (1-t)^{-5/6}.$$

Thus the derivative of the actual function is

$$H'(x) = \frac{1}{12\pi} \int_0^1 t^{5/6} (1-t)^{-5/6} (1-xt)^{-7/6} dt. \quad (64)$$

Take $x = 1 - \delta$ with $0 < \delta \leq 1/2$, first set $r_0 = 1 - t$ and then $r_0 = \delta v$. This gives the exact equation

$$\delta H'(1 - \delta) = \frac{1}{12\pi} \int_0^{1/\delta} (1 - \delta v)^{5/6} v^{-5/6} \{1 + (1 - \delta)v\}^{-7/6} dv. \quad (65)$$

Extend this integrand by zero for $v > 1/\delta$. It is bounded by $v^{-5/6}(1 + v/2)^{-7/6}$, which is integrable: it is $O(v^{-5/6})$ at zero and $O(v^{-2})$ at infinity. Dominated convergence and the substitution $w = v/(1 + v)$ now give

$$\int_0^\infty v^{-5/6} (1 + v)^{-7/6} dv = \int_0^1 w^{-5/6} dw = 6.$$

Consequently

$$\lim_{\delta \downarrow 0} \delta H'(1 - \delta) = \frac{1}{2\pi}. \quad (66)$$

The undifferentiated integral has a sufficient elementary bound of its own: $1 - (1 - \delta)t \geq \delta$ for $0 \leq t \leq 1$. By (62) and (63),

$$0 < H(1 - \delta) \leq \delta^{-1/6}, \quad 0 < \delta H(1 - \delta) \leq \delta^{5/6} \longrightarrow 0 \quad (\delta \downarrow 0). \quad (67)$$

For $0 < q < 1$ define the Euler Wronskian

$$\mathcal{W}(q) = g(q)\theta Z(q) - (\theta g(q))Z(q).$$

Equation (43) gives

$$\theta \mathcal{W} = \frac{q}{2(1-q)} \mathcal{W}, \quad \frac{d}{dq} \{\sqrt{1-q} \mathcal{W}(q)\} = 0.$$

Its constant is fixed by (66): as $q \downarrow 0$, $p(q) = q/4 + O(q^2)$ and

$$\frac{q p'(q)}{p(q)} = \frac{1 + \sqrt{1-q}}{2\sqrt{1-q}} \longrightarrow 1, \quad \theta Z(q) = -q p'(q) H'(1 - p(q)) \longrightarrow -\frac{1}{2\pi}.$$

Also $g(q) \rightarrow 1$ and $\theta g(q) = q p'(q) H'(p(q)) = O(p(q))$: H' is bounded near zero and $q/p(q) \rightarrow 4$. The independent bound (67) therefore gives $(\theta g(q))Z(q) = O(p(q)^{5/6}) \rightarrow 0$. We have proved

$$g(q)\theta Z(q) - (\theta g(q))Z(q) = -\frac{1}{2\pi\sqrt{1-q}}. \quad (68)$$

In particular the derivative normalization comes directly from the differentiated integral and its integrable bound, not from differentiating a remainder in a bare logarithmic expansion.

8.4 The actual contact at $h = 64$

Put $u = 27/125$ and $s_u = \sqrt{1-u} = 7\sqrt{10}/25$. Substitution in (47) gives

$$\begin{aligned} q_1(64) &= q_2(64) = u, & L(64) &= \sqrt{2}, \\ v_1(64) &= s_u, & v_2(64) &= -s_u, \\ p_1(64) &= p(u), & p_2(64) &= 1 - p(u). \end{aligned} \tag{69}$$

Therefore the actual transformation already proves

$$Z(u) = \sqrt{2}g(u). \tag{70}$$

On a neighborhood of $h = 64$ contained in $(8, 512)$ the positive-root definition in (41) satisfies $p_1(h) = p(q_1(h))$ and $p_2(h) = 1 - p(q_2(h))$. This follows from $4p_i(1-p_i) = q_i$, $v_1 > 0$ and $v_2 < 0$, which are also consequences of (50). We may consequently differentiate the actual equality $Z(q_2(h)) = L(h)g(q_1(h))$ on that neighborhood. The three logarithmic derivatives are

$$\frac{q'_1}{q_1} = \frac{1}{h} - \frac{3}{h+16}, \quad \frac{q'_2}{q_2} = \frac{2}{h} - \frac{3}{h+256}, \quad \frac{L'}{L} = -\frac{60}{(h+16)(h+256)},$$

so at $h = 64$ their values are $-7/320$, $7/320$ and $-3/1280$. The differentiated identity is thus

$$\frac{7}{320}\theta Z(u) = \sqrt{2}\left\{-\frac{3}{1280}g(u) - \frac{7}{320}\theta g(u)\right\},$$

or, equivalently,

$$\theta Z(u) = -\sqrt{2}\theta g(u) - \frac{3\sqrt{2}}{28}g(u). \tag{71}$$

This relation uses the ordinary derivative of an identity on an open interval, with all moving parameters and the derivative of L retained.

Finally insert (70) and (71) into (68), and use $F = g^2$ and $\theta F = 2g\theta g$ from (46). It follows that

$$-\sqrt{2}\left\{\theta F(u) + \frac{3}{28}F(u)\right\} = -\frac{1}{2\pi s_u}.$$

Since $s_u = 7\sqrt{10}/25$, this is the required native certificate

$$\boxed{3F\left(\frac{27}{125}\right) + 28\theta F\left(\frac{27}{125}\right) = \frac{5\sqrt{5}}{\pi}}. \tag{72}$$

9 Completion of the series evaluation

Equation (72) substituted in (39) gives

$$\mathfrak{b}(0+) = \frac{(5\sqrt{5}/\pi)^2}{4500} = \frac{1}{36\pi^2}.$$

Together with (35), this gives $\mathcal{E}(0+) = 1/(36\pi^2)$. The upper limit is (34), and the escape singularity has been removed. On every closed subinterval of $(0,1)$ the extended primitive satisfies (28). The loaded radial integrand is continuous and bounded on $[0,1]$, by its uniformly convergent polynomial-geometric series. Applying the fundamental theorem of calculus first away from the endpoints and then taking their limits yields

$$\int_0^1 Q(\theta_\lambda)h(\rho(1-U^2)^3) dU = 3375\{\mathcal{E}(0+) - \mathcal{E}(1-)\} = \frac{3375}{36\pi^2} = \frac{375}{4\pi^2}.$$

Finally (8) identifies this entire integral with the literal factorial series in Theorem 1.1, proving the theorem.

A A finite rational certificate for the source row

This appendix supplies the rational identities used in the contour comparison. In particular, it includes an actual primitive, rather than a rank computation or a numerical fit. Everything in the certificate is defined below over rational function fields. The contour's analytic boundary and normalization are the statements proved in the main text; finite algebra by itself would not establish those statements.

Write

$$\rho = \frac{729}{15625}, \quad x = 1 - U^2, \quad \lambda = \rho x^3, \quad b = \frac{\lambda}{s}, \quad \tau = 2\pi i.$$

All coefficient rows act on solution columns. Thus their covariant differential has a *plus* sign. Tensor coordinates are ordered (00, 01, 02, 10, 11, 12, 20, 21, 22).

A.1 The complete source coefficient

The ordinary elliptic symmetric-square state is $(y^2, 2yy', y'^2)^\top$; its third entry is not the second derivative of its first entry. Its connection and Euler gauge are

$$A(q) = \begin{pmatrix} 0 & 1 & 0 \\ 2c(q) & -a(q) & 2 \\ 0 & c(q) & -2a(q) \end{pmatrix}, \quad a(q) = \frac{1-3q/2}{q(1-q)}, \quad c(q) = \frac{5}{144q(1-q)},$$

$$G(q) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & q & 0 \\ 5q & q^2 & 2q^2 \end{pmatrix}, \quad N(q) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 5q & 23q & 3q \end{pmatrix}.$$

They satisfy $G' + GA = (N/q)G$. The solution-state form and its useful product with the gauge are

$$J_C(q) = \begin{pmatrix} -5q/36 & -q/2 & 1-q \\ -q/2 & q-1 & 0 \\ 1-q & 0 & 0 \end{pmatrix},$$

$$T(q) = J_C(q)G(q) = \begin{pmatrix} -5q/72 & 0 & 2q^2(1-q) \\ -q/2 & q(q-1) & 0 \\ 1-q & 0 & 0 \end{pmatrix}.$$

Here are the coefficient polynomials of the literal rational source. Put

$$\begin{aligned} d &= x^4(25-9x)(99x-50)^2, & K_2 &= 81x^2 - 198x + 80, \\ K_3 &= 243x^3 - 1053x^2 + 1266x - 400, & K_6 &= 59049x^6 - 511758x^5 + 1724085x^4 \\ & & & - 2860596x^3 + 2441988x^2 - 1011200x + 160000, \\ k &= \frac{1953125}{209952}, & j &= \frac{244140625}{944784}, \\ v &= \frac{3906250}{6561}, & c_* &= \frac{30517578125}{8503056}. \end{aligned}$$

Then

$$L_P = \begin{pmatrix} 0 & 0 & -k(9x-25)K_2K_3 \\ 0 & 0 & j(9x-25)K_2 \\ k(9x-25)K_2K_3 & -j(9x-25)K_2 & 0 \end{pmatrix},$$

$$R_P = \begin{pmatrix} 0 & 0 & kK_6 \\ 0 & vx(99x-50) & -jK_3 \\ kK_6 & -jK_3 & c_* \end{pmatrix}.$$

For compact notation put $L = UL_P/d$ and $R = R_P/d$. Define

$$m = \frac{27x(80 - 99x + 27x^2)}{1000}, \quad \ell_0 = \frac{27x(9x - 16)}{1000}, \quad \ell = U\ell_0, \quad M = 25 - 9x,$$

$$z = s - m, \quad f = z^2 - \ell^2 M = s^2 - 2ms + \lambda.$$

The entire rational coefficient one-form is

$$\beta_N = \frac{L df + R(2\ell M dz - 2zM d\ell - z\ell dM)}{f} = T_U dU + UT_s ds, \quad (73)$$

where, with primes denoting ∂_x at fixed s ,

$$T_s = \frac{2L_P(s - m) + 2\ell_0 MR_P}{df}, \quad (74)$$

$$T_U = \frac{-2(1 - x)L_P f_x - 2R_P\{-2\ell_0(1 - x)Mm' + (s - m)[(\ell_0 - 2(1 - x)\ell'_0)M + 9(1 - x)\ell_0]\}}{df}. \quad (75)$$

Thus the correction has not been omitted or absorbed into an unspecified boundary term. The relation to the main coefficients is $B_s = UT_s$ and $B_U = T_U$. Taking their covariant derivatives gives

$$\begin{aligned} \mathbf{K} = & -T_s + 2(1 - x)\partial_x T_s - \frac{6(1 - x)b}{x} T_s \mathbf{A}(b)^\top \\ & + \partial_s T_U - \mathbf{A}(s)T_U + \frac{b}{s} T_U \mathbf{A}(b)^\top, \end{aligned} \quad (76)$$

$$\mathbf{q} = \text{flatten}(T(s)\mathbf{K}T(b)^\top). \quad (77)$$

In particular, the source row against the native product Euler state is $\mathbf{q} dU \wedge ds$. Thus $\mathbf{q} = \text{flatten}(\mathbf{S})$ in the notation of the main text. Formula (76) is simply $-\nabla_U B_s + \nabla_s B_U$, with $x_U = -2U$ and $b_U = -6Ub/x$; these identities fix both its sign and its chain-rule terms.

A.2 Angular normal form and its six-state action

Let

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 5/72 & 23/36 & 3/2 \end{pmatrix}, \quad E = \mathbf{e}_2^\top \mathbf{w},$$

Here $\mathbf{e}_2 = (0, 0, 1)$ is the standard row, as in the main text. Over $\mathbb{Q}(\lambda)(s)$,

$$\mathcal{M}_s = \frac{N(s) \otimes I - I \otimes N(\lambda/s)}{s} = \frac{R_0}{s} + \frac{R_1}{s - 1} + \frac{R_\lambda}{s - \lambda},$$

$$R_0 = J \otimes I - I \otimes (J - E), \quad R_1 = -E \otimes I, \quad R_\lambda = -I \otimes E.$$

For a six-row $v = (a_0, a_1, a_2, b_0, b_1, b_2) = (a, b_*)$, define

$$\text{NF}(v) = \frac{\mathbf{w} \otimes a}{s - 1} + \frac{b_* \otimes \mathbf{w}}{s - \lambda}. \quad (78)$$

The six standard coordinate rows under this formula are the six angular basis representatives. Their induced D_λ action is

$$B_6(\lambda) = \begin{pmatrix} J + \frac{\lambda E}{1 - \lambda} & -\frac{\lambda E}{1 - \lambda} \\ -\frac{E}{1 - \lambda} & J + \frac{\lambda E}{1 - \lambda} \end{pmatrix}. \quad (79)$$

This assertion includes an explicit angular primitive. For a constant row $v = (a, b_*)$, put $p_v = -s(b_* \otimes \mathbf{w})/(s - \lambda)$. Entrywise,

$$\lambda \partial_\lambda \text{NF}(v) + \text{NF}(v)(I \otimes N(\lambda/s)) - \text{NF}(vB_6) = \partial_s p_v + p_v \mathcal{M}_s. \quad (80)$$

For the first three basis rows $p_v = 0$.

The coefficient one-form for the anchor is $\omega_0 = e_{00} ds/s$, whose normalized contour integral is the scalar function h . If $\eta_0 = e_{00} R_0^{-1}$, its exact angular identity is $e_{00}/s = \text{NF}(c_0) + \eta_0 \mathcal{M}_s$. The coordinate rows of $\omega_0, D_\lambda \omega_0, \dots, D_\lambda^5 \omega_0$ form the following full matrix:

$$C = \begin{pmatrix} 49456/125 & 153648/125 & 113472/125 & 72/5 & -3312/25 & 113472/125 \\ -1576/25 & -4608/25 & -3312/25 & 0 & 72/5 & -3312/25 \\ 46/5 & 108/5 & 72/5 & 0 & 0 & 72/5 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}. \quad (81)$$

Writing its rows as $c_0, \dots, c_5$, one has $c_j B_6 = c_{j+1}$ for $0 \leq j < 5$ and $\det C = 373248/125$. These are identities in the full six-state system.

For completeness, the angular elimination used below is finite and unambiguous. At a finite puncture p , a leading pole row $v/(s - p)^m$, $m \geq 2$, is removed by subtracting the covariant derivative of

$$\frac{v(R_p - (m - 1)I)^{-1}}{(s - p)^{m-1}}.$$

At infinity a polynomial leading row vs^d is removed by subtracting the derivative of $v((d + 1)I + R_0 + R_1 + R_\lambda)^{-1}s^{d+1}$. The eigenvalues of R_0 are $1/6, 1/2, 5/6$, each repeated three times; those of R_1, R_λ are 0 and $-3/2$; those of $R_0 + R_1 + R_\lambda$ are $-1/6, -1/2, -5/6$, each repeated three times. Every inverse just used therefore exists. A final constant primitive removes the simple zero residue because R_0 is invertible. Finite primitives decay at infinity and create only simple poles at other finite punctures; polynomial primitives also create only simple finite poles. Thus each step strictly lowers the relevant higher pole or polynomial degree and the procedure terminates. The same leading-coefficient argument proves uniqueness of the remaining form $v_1/(s - 1) + v_\lambda/(s - \lambda)$.

Applying this stated procedure to (77) gives a rational row η_S and six-row $G(x)$ satisfying

$$\mathbf{q} = \text{NF}(G) + \partial_s \eta_S + \eta_S \mathcal{M}_s. \quad (82)$$

The listing below constructs and retains the primitive; its variable **eta**, divided by the coefficient-field scaling **scale** $= x^5(9x - 25)(99x - 50)^3$, is η_S . It verifies this entire nine-entry identity and the factorization of both residues in (78). No unspecified angular witness is needed.

A.3 The outer primitive and all its coefficients

Use the six-coordinate solution column $\mathcal{V}$ obtained by multiplying $\mathbf{v}(\rho x^3)$ by $U = \sqrt{1 - x}$. Thus $\mathcal{V} = U \mathbf{v}(\rho x^3)$, and its ordinary x -connection is

$$M_x = \frac{3B_6(\rho x^3)}{x} - \frac{I}{2(1 - x)}.$$

Since $dU = -dx/(2U)$, the source row and three normalized period rows in this frame are

$$r_S = -\frac{G}{2(1 - x)}, \quad H_j = -\frac{c_j + 6c_{j+1}}{2(1 - x)}, \quad 0 \leq j \leq 2.$$

Here D_λ denotes the coefficient action corresponding to θ_λ ; the full factor $1 + 6D_\lambda$ is retained. Define

$$(d_0, d_1, d_2) = \left(\frac{1}{375}, \frac{14}{375}, \frac{532}{3375} \right).$$

An actual rational primitive is

$$\begin{aligned} \xi(x) &= \sum_{k=1}^{15} A_k \phi_k(x), \\ (\phi_1, \dots, \phi_6) &= (x^{-1}, x^{-2}, x^{-3}, x^{-4}, (99x - 50)^{-1}, (99x - 50)^{-2}), \\ (\phi_7, \dots, \phi_{15}) &= (1, x, \dots, x^8). \end{aligned} \tag{83}$$

All ninety rational entries of the rows A_k appear literally in `COEFFICIENTS` in the listing. No coefficient is a variable to be solved for. With that witness the exact identity is

$$r_S - d_0 H_0 - d_1 H_1 - d_2 H_2 = \xi' + \xi M_x. \tag{84}$$

It can be checked by clearing denominators in six rational functions. Neither existence of a primitive nor a claim about a computed rank is a premise of this equality.

The angular primitive enters the two-form identity with its wedge sign: the exact one-form $-\eta_S dU$, paired with the product solution column, contributes $(\partial_s \eta_S + \eta_S \mathcal{M}_s) dU \wedge ds$. The outer term is the ordinary derivative of $\xi \mathcal{V}$. Hence (82) and (84) are the actual primitives required for the comparison. Their poles at $99x - 50 = 0$ remain present; the contour truncation and endpoint argument in the main text justify applying the identities to the selected literal contour.

A.4 A complete exact verification recipe

The following Python listing uses only exact SymPy arithmetic. It is a self-contained transcription of the finite identities: it reads no previous certificate or saved output, executes no solver, and contains the complete coefficient witness. Equality of rational functions is tested by exact polynomial cancellation. Its source is obtained from (73), not inserted as a fitted six-row. During angular reduction, multiplying by $x^5(9x - 25)(99x - 50)^3$ is only a coefficient-field scaling and is undone in the result. For generic x and $\lambda \neq 0, 1$, all steps are defined; the resulting rational identities consequently hold throughout their domains by clearing denominators.

```

"""Standalone exact transcription check for the printed rational appendix.

This file supplies its own literal inputs and never reads or executes an old
certificate, solver, or output. No period, target fit, or rank is an input.
The printed 90 coefficients constitute a witness, not unknowns to be solved.
"""

import sympy as S
from sympy.polys.fields import field
import time

started = time.perf_counter()
X, Z, LAM = S.symbols('x s lambda')
rho = S.Rational(729, 15625)
I = S.eye(3)
J = S.Matrix([[0, 1, 0], [0, 0, 1], [0, 0, 0]])
w = S.Matrix([[S.Rational(5,72), S.Rational(23,36), S.Rational(3,2)]])
E = S.zeros(3)
E[2,:] = w
R0 = S.kronecker_product(J,I) - S.kronecker_product(I,J-E)
R1 = -S.kronecker_product(E,I)
R1 = -S.kronecker_product(I,E)
Rt = R0 + R1 + R1
B6 = (J + LAM*E/(1-LAM)).row_join(-LAM*E/(1-LAM)).col_join(
    (-E/(1-LAM)).row_join(J + LAM*E/(1-LAM)))
C = S.Matrix([
    [S.Rational(49456,125), S.Rational(153648,125), S.Rational(113472,125),
     S.Rational(72,5), -S.Rational(3312,25), S.Rational(113472,125)],

```

```

[-S.Rational(1576,25),-S.Rational(4608,25),-S.Rational(3312,25),
 0,S.Rational(72,5),-S.Rational(3312,25)],
[S.Rational(46,5),S.Rational(108,5),S.Rational(72,5),0,0,S.Rational(72,5)],
[-1,0,0,0,0,0],[0,-1,0,0,0,0],[0,0,-1,0,0,0]])

def cancel_matrix(A):
    return A.applyfunc(S.cancel)

def nf(v):
    return (S.kronecker_product(w,v[:,3:3])/(Z-1)
            + S.kronecker_product(v[:,3:],w)/(Z-LAM))

N = lambda q: J + q*E/(1-q)
gauge = S.Matrix([[1,0,0],[0,LAM,0],
                  [5*LAM/(72*(1-LAM)),LAM**2/(2*(1-LAM)),2*LAM**2]])
pq,cq = (1-S.Rational(3,2)*LAM)/(LAM*(1-LAM)),5/(144*LAM*(1-LAM))
ao = S.Matrix([[0,1,0],[2*cq,-pq,2],[0,cq,-2*pq]])
jc = S.Matrix([[[-5*LAM/36,-LAM/2,1-LAM],[-LAM/2,LAM-1,0],[1-LAM,0,0]])
tq = S.Matrix([[[-5*LAM/72,0,2*LAM**2*(1-LAM)],
                [-LAM/2,LAM*(LAM-1),0],[1-LAM,0,0]])
assert cancel_matrix(gauge.diff(LAM)+gauge*ao-N(LAM)*gauge/LAM) == S.zeros(3)
assert cancel_matrix(jc*gauge-tq) == S.zeros(3)
Ms = S.kronecker_product(N(Z),I)/Z - S.kronecker_product(I,N(LAM/Z))/Z
Ml = S.kronecker_product(I,N(LAM/Z))
assert cancel_matrix(Ms-R0/Z-R1/(Z-1)-Rl/(Z-LAM)) == S.zeros(9)
for j in range(6):
    v = S.eye(6)[j,: ]
    p = S.zeros(1,9) if j < 3 else -Z*S.kronecker_product(v[:,3:],w)/(Z-LAM)
    assert cancel_matrix(LAM*nf(v).diff(LAM)+nf(v)*Ml-nf(v*B6)
                        -p.diff(Z)-p*Ms) == S.zeros(1,9)
e00 = S.eye(9)[0,: ]
anchor = e00*R0.inv()
assert cancel_matrix(e00/Z-nf(C[0,:])-anchor*Ms) == S.zeros(1,9)
for j in range(5):
    assert cancel_matrix(C[j,:]*B6-C[j+1,:]) == S.zeros(1,6)
assert C.det() == S.Rational(373248,125)

# The rational beta source, with beta_N = T_U dU + U*T_s ds.
x, s, lam = X, Z, rho*X**3
m = 27*x*(80-99*x+27*x**2)/1000
a = 27*x*(9*x-16)/1000
M = 25-9*x
D = x**4*M*(99*x-50)**2
f = s**2-2*m*s+lam
a2 = 81*x**2-198*x+80
a3 = 243*x**3-1053*x**2+1266*x-400
a6 = (59049*x**6-511758*x**5+1724085*x**4-2860596*x**3
      +2441988*x**2-1011200*x+160000)
k,jj,vv,cc = map(S.Rational, ['1953125/209952','244140625/944784',
                               '3906250/6561','30517578125/8503056'])
LP = S.Matrix([[0,0,-k*(9*x-25)*a2*a3],
               [0,0,jj*(9*x-25)*a2],
               [k*(9*x-25)*a2*a3,-jj*(9*x-25)*a2,0]])
RP = S.Matrix([[0,0,k*a6],[0,vv*x*(99*x-50),-jj*a3],
               [k*a6,-jj*a3,cc]])

# These temporary matrices are the displayed beta numerators.
# Their common scalar denominator is D*f.
Ts = cancel_matrix(2*LP*(s-m)+2*a*M*RP)
TU = cancel_matrix((-2*(1-x)*LP*S.diff(f,x)-2*RP*(
    -2*a*(1-x)*M*S.diff(m,x)+(s-m)*(
    (a-2*(1-x)*S.diff(a,x))*M+9*(1-x)*a))))

# The beta numerators are polynomials in s over Q(x).
# Clear the known connection denominator, divide by f^2 exactly, and only
# then create rational rows. This preserves all nine source entries.
F, x = field('x',S.QQ)
K, s = field('s',F.to_domain())
z = K.zero
lam = rho*x**3
lift = K.ground_new
parse = lambda q: K.from_expr(q)
constant = lambda A: [[parse(A[i,j]) for j in range(A.cols)] for i in range(A.rows)]

```

```

ts_fraction,tu_fraction = constant(Ts),constant(TU)
assert all(q.denom.degree() == 0 for row in ts_fraction+tu_fraction for q in row)
ts = [[q.numer.mul_ground(F.one/q.denom.LC) for q in row] for row in ts_fraction]
tu = [[q.numer.mul_ground(F.one/q.denom.LC) for q in row] for row in tu_fraction]
P = K.ring
sp = s.numer
pp = P.one
zz = P.zero
u = 1-x
mfield = F.from_expr(m)
dlogD = F.from_expr(S.diff(D,X)/D)
fpoly = sp**2-2*mfield*sp+lam

def dx_polynomial(p):
    return P.from_dict({e:c.diff(x) for e,c in p.items()})

fx,fs = dx_polynomial(fpoly),fpoly.diff(sp)
angular_den = sp**2*(1-sp)*(sp-lam)
asbar = [[zz,sp*(1-sp),zz],
          [S.Rational(5,72)*pp,-pp+S.Rational(3,2)*sp,2*sp*(1-sp)],
          [zz,S.Rational(5,144)*pp,-2*pp+3*sp]]
cbbar = [[zz,lam*(sp-lam),zz],
          [S.Rational(5,72)*sp**2,-sp**2+S.Rational(3,2)*lam*sp,2*lam*(sp-lam)],
          [zz,S.Rational(5,144)*sp**2,-2*sp**2+3*lam*sp]]
scale = lift(x**5*(9*x-25)*(99*x-50)**3)
scale_over_D = -x*(99*x-50)
alpha = [[z]*3 for _ in range(3)]
for i in range(3):
    for j in range(3):
        scalar_part = (-ts[i][j]*fpoly
                        +2*u*(dx_polynomial(ts[i][j])*fpoly-ts[i][j]*fx
                              -dlogD*ts[i][j]*fpoly)
                        +tu[i][j].diff(sp)*fpoly-tu[i][j]*fs)
        numerator = (angular_den*scalar_part
                     -sp*(sp-lam)*sum((asbar[i][k]*tu[k][j] for k in range(3)),zz)*fpoly
                     +(1-sp)*sum((tu[i][k]*cbbar[j][k] for k in range(3)),zz)*fpoly
                     -(6*u/x)*sp*(1-sp)*sum((ts[i][k]*cbbar[j][k] for k in range(3)),zz)*fpoly)
        quotient,remainder = numerator.div(fpoly**2)
        assert remainder == zz
        alpha[i][j] = K.new(quotient.mul_ground(scale_over_D),angular_den)
print('All nine source entries pass exact zero-remainder f^2 division.', flush=True)

def metric_gauge(q):
    return [[-5*q/72,z,2*q*(1-q)],
            [-q/2,q*(q-1),z],[1-q,z,z]]

b = lift(lam)/s
left,right = metric_gauge(s),metric_gauge(b)
Q = [[sum((left[i][a]*alpha[a][c]*right[j][c]
           for a in range(3) for c in range(3))),z)
      for j in range(3)] for i in range(3)]
print('Source transcribed from the complete beta formulas.', flush=True)

r0,r1,r1 = map(constant,[R0,R1,R1])
ms = [[r0[i][j]/s+r1[i][j]/(s-1)+r1[i][j]/(s-lift(lam))
        for j in range(9)] for i in range(9)]

def row_times(v,A):
    return [sum((v[i]*A[i][j] for i in range(len(v))),z)
            for j in range(len(A[0]))]

def covariant(v):
    return [v[j].diff(s)+q for j,q in enumerate(row_times(v,ms))]

def evaluate(q,a):
    def ev(p):
        return sum((c*(a**e[0] if e[0] else F.one) for e,c in p.items()),F.zero)
    return lift(ev(q.numer)/ev(q.denom))

def pole_order(q,a):
    p,linear,n = q.denom,(s-lift(a)).numer,0
    while q:
        quotient,remainder = p.div(linear)

```

```

        if remainder:
            break
        p,n = quotient,n+1
    return n

q = [Q[i][j] for i in range(3) for j in range(3)]
r, eta = q[:,z]*9

def subtract(p):
    global r,eta
    exact = covariant(p)
    r = [r[j]-exact[j] for j in range(9)]
    eta = [eta[j]+p[j] for j in range(9)]

for a,residue in [(F.one,R1),(lam,R1),(F.zero,R0)]:
    while True:
        order = max(pole_order(t,a) for t in r)
        if order < 2:
            break
        lead = [evaluate(t*(s-lift(a))**order,a) for t in r]
        inv = constant((residue-(order-1)*S.eye(9)).inv())
        subtract([t/(s-lift(a))**(order-1) for t in row_times(lead,inv)])
    while True:
        polynomials = [t.numer.div(t.denom)[0] for t in r]
        degree = max(p.degree(0) if p else -1 for p in polynomials)
        if degree < 0:
            break
        lead = [lift(p.get((degree,)),F.zero)) for p in polynomials]
        inv = constant(((degree+1)*S.eye(9)+Rt).inv())
        subtract([t*s**(degree+1) for t in row_times(lead,inv)])
    subtract(row_times([evaluate(t*s,F.zero) for t in r],constant(R0.inv())))
    u = [evaluate(t*(s-1),F.one) for t in r]
    v = [evaluate(t*(s-lift(lam)),lam) for t in r]
    normal = [u[j]/(s-1)+v[j]/(s-lift(lam)) for j in range(9)]
    assert all(not (r[j]-normal[j]) for j in range(9))
    exact = covariant(eta)
    assert all(not (q[j]-normal[j]-exact[j]) for j in range(9))
    g = [u[6+j]/parse(w[0,2])/scale for j in range(3)]
    g += [v[3*j+2]/parse(w[0,2])/scale for j in range(3)]
    assert all(not (u[3*i+j]-parse(w[0,i])*g[j]*scale)
                and not (v[3*i+j]-g[3+i]*parse(w[0,j])*scale)
                for i in range(3) for j in range(3))
    print('Angular source identity established with its rational primitive.', flush=True)

# Literal 15 by 6 coefficient table: basis order is specified below.
COEFFICIENTS = '''
105800/459459 867560/459459 103229200/459459 -169280/51051 -1388096/51051 -2268352/51051
0 0 -472485400/459459 0 0 0
0 0 338560000/196911 0 0 0
0 0 -4232000000/4135131 0 0 0
-78890805485931260/886402053948411 -1227125311357021190/886402053948411
-70427029255928891500/6204814377638877 -607873567055897705440/74327471429736107583
-298973396640778498112/8258607936637345287 -145365088004028281168/635277533587488099
-231954922267076000/229807939912551 -331597299956488000/76602646637517 -108998157631733600/5892511279809
-39913346954892419000/917623104070816143 28252939480465998800/101958122674535127
-14427468858797595200/5997536627913831
-44128649359258592/31024071888194385 623049674458568524/31024071888194385
6660655461825680912/31024071888194385 57598947351747925330/10618210204248015369
370356988491186478316/8258607936637345287 422172179336846381488/5899005669026675205
7410960991604618219/7834361587927875000 -2244024947258679397/230422399644937500
-836844458835533518/75330399883921875 -1674067760127596905471/469239087308940073125
-7571431721389433993041/260688381838300040625 -60217890176084166521579/1303441909191500203125
-1014783336398633/79134965534625000 99347762374288343/39567482767312500 226386615454772638/9891870691828125
1399948922915894608/4739788760696364375 6901571092120745041/2633215978164646875
59461141531221275693/13166079890823234375
-1051211136/19184234069 -440067361152/479605851725 -18358156683264/2398029258625
1031881871061389312/1196916353711203125 258130738760714016/40711440602421875
20750847913747165488/1994860589518671875
1029377673/34880425580 430927242561/872010639500 4494215281788/1090013299375
-250759381828805733/558002962103125000 -14957478836244716238/4533774067087890625
-24572653010256580956/4533774067087890625
-16224867/3170947780 -6792198219/79273694500 -70837018452/99092118125
300193629643629927/3297290230609375000 1380547922050429344/2060806394130859375

```

```

2261266906791518928/2060806394130859375
0 0 0 -223303109482368/7493841433203125 -8431993935111168/37469207166015625
-13467359214383616/37469207166015625
0 0 0 218665144746549/13625166242187500 2064215740879206/17031457802734375
3296911156760547/17031457802734375
0 0 0 -3446560950471/1238651476562500 -32535799768674/1548314345703125 -51965324712513/1548314345703125
'''
coefficients = [[F.from_expr(S.Rational(t)) for t in line.split()]
                 for line in COEFFICIENTS.strip().splitlines()]
assert len(coefficients) == 15 and all(len(row) == 6 for row in coefficients)
basis = [x**(-p) for p in range(1,5)]
basis += [(99*x-50)**(-p) for p in range(1,3)] + [x**p for p in range(9)]
xi = [sum((coefficients[k][j]*basis[k] for k in range(15)),F.zero)
      for j in range(6)]
mx = [[3*F.from_expr(B6[i,j].subs(LAM,lam.as_expr()))/x
      -(F.one/(2*(1-x)) if i == j else F.zero)
      for j in range(6)] for i in range(6)]
d = list(map(S.Rational,['1/375','14/375','532/3375']))
h = [[-F.from_expr(C[k,j]+6*C[k+1,j])/(2*(1-x))
      for j in range(6)] for k in range(3)]
for j in range(6):
    residual = (-g[j].as_expr()/(2*(1-x))
               -sum(d[k]*h[k][j].as_expr() for k in range(3))
               -xi[j].diff(x).as_expr()
               -sum((xi[i]*mx[i][j]).as_expr() for i in range(6)))
    assert S.cancel(residual) == 0
assert [-81*t for t in d] == list(map(S.Rational,['-27/125','-378/125','-1596/125']))
print('All finite printed identities hold over the rational function fields.')
print('Elapsed seconds:', round(time.perf_counter()-started,3))

```

B Provenance and review boundary

The finite rational inputs were transcribed from the frozen Row12 proof package associated with author commit `a17703b4e69a5d879d71ebd37fcaba32af45be0c` and strict-audit package `7b71f2c0f1c72798dda3e8759da4465b489366c0`. In that package the relevant inputs are the source-lift proof and `s03/LIFT_CHECK.output.json`, the six-coordinate proof `s08/PROOF.md`, and the complete outer coefficients `s14/OUTER_ROW_CHECK.output.json`. The finite formulas and verification procedure used in this document are printed in Appendix A, so those historical files are provenance references rather than additional compilation inputs.

The analytic scalar route was organized in `SCALAR_INTEGRAL_ROUTE.md` and checked together with the literal coefficient and moment bridge in the additional scalar review performed 7 October 2026. The signed Gauss calculation uses the corrected actual derivatives recorded in the completed signed-Gauss Lean support package; the earlier unchecked draft is not a premise of this proof. The document-level review, its checked source version and any corrections are recorded next to this source in `DOCUMENT_REVIEW.md`.

The evidential status is [ARG] for the analytic derivation and [ONE] for the exact document-transcription checks. Human acceptance is Pending. The connected Lean formalization was completed on 7 October 2026. Its unconditional declarations are `Row12.row12_hasSum`, `Row12.row12_summable` and `Row12.row12_tsum`. The final statement and axiom outputs, all five verification checks and the fresh full-result review are retained in the separate `C5/row12_lean_2026-10-06` package. That formal result covers exactly the scalar equality proved here.